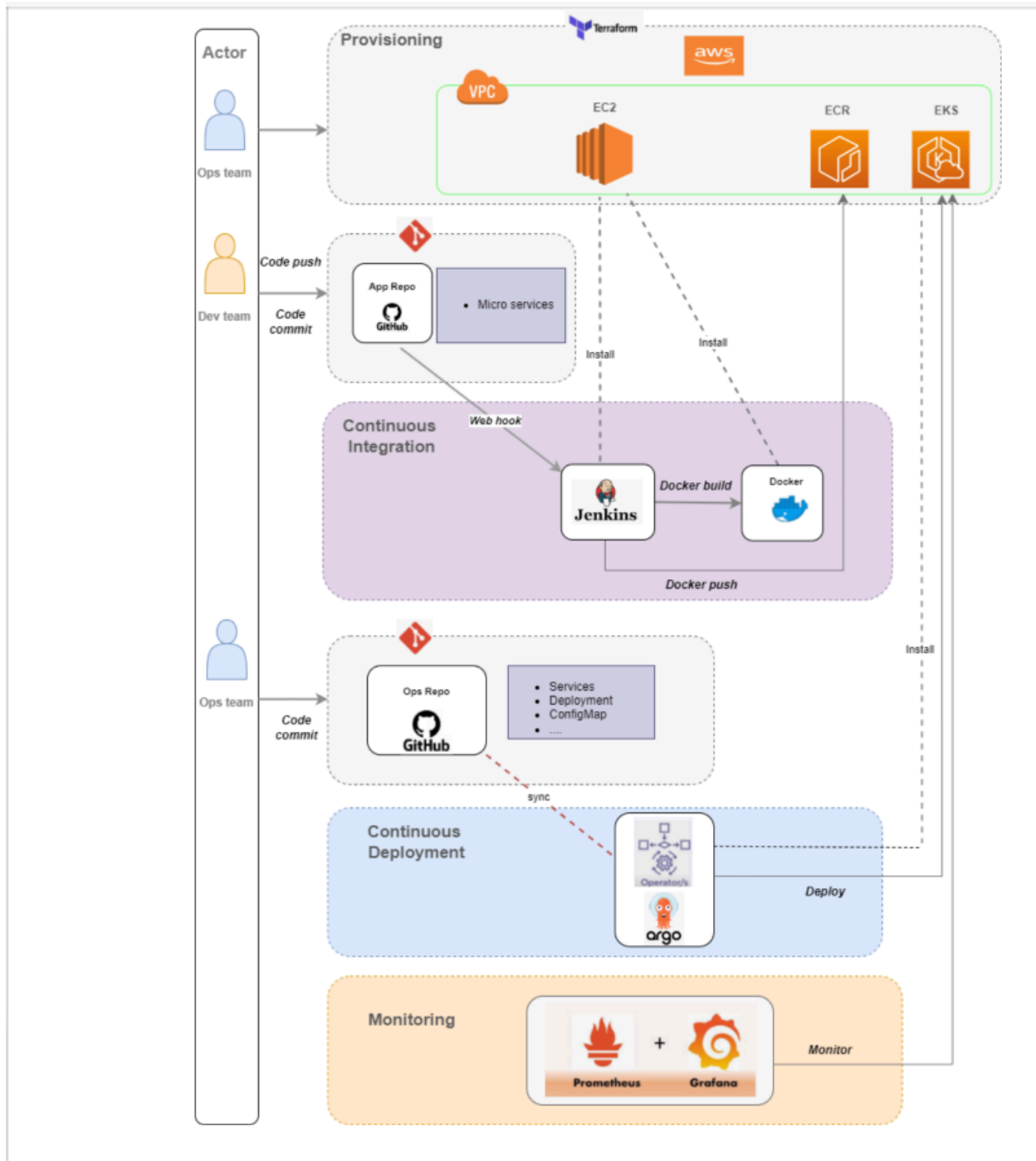


Setting up a CI/CD Pipeline and deploying applications on AWS EKS



1. Source Code Management

- msa application repository:

https://github.com/nvvuong2000/sd4413_msa

- aws infrastructure repository:

https://github.com/nvvuong2000/sd4413_azure_infrastructure

2. Use Terraform to provision VPC, EC2, ECR and EKS on AWS

```
vuongnguyenv@VNNOT02236:~/sd4413_azure_infrastructure/aws$ terraform init

Initializing the backend...

Initializing provider plugins...
- Finding hashicorp/helm versions matching "2.15.0"...
- Finding hashicorp/aws versions matching "5.63.0"...
- Installing hashicorp/helm v2.15.0...
- Installed hashicorp/helm v2.15.0 (signed by HashiCorp)
- Installing hashicorp/aws v5.63.0...
- Installed hashicorp/aws v5.63.0 (signed by HashiCorp)

Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
vuongnguyenv@VNNOT02236:~/sd4413_azure_infrastructure/aws$ terraform validate
```

```

vuongnguyenv@VNNOT02236:~/sd4413_azure_infrastructure/aws$ terraform apply -var-file="dev.tfvars" -auto-approve
data.aws_iam_role.eks_service_role: Reading...

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following
symbols:
+ create

Terraform planned the following actions, but then encountered a problem:

# aws_ecr_repository.backend will be created
+ resource "aws_ecr_repository" "backend" {
+   arn                = (known after apply)
+   id                 = (known after apply)
+   image_tag_mutability = "MUTABLE"
+   name                = "sd4413-practical-devops"
+   registry_id         = (known after apply)
+   repository_url      = (known after apply)
+   tags_all            = {
+     "Course" = "sd4413-devops-for-dev"
+   }
+ }

# aws_ecr_repository.frontend will be created
+ resource "aws_ecr_repository" "frontend" {
+   arn                = (known after apply)
+   id                 = (known after apply)
+   image_tag_mutability = "MUTABLE"
+   name                = "sd4413-practical-devops"
+   registry_id         = (known after apply)
+   repository_url      = (known after apply)
+   tags_all            = {
+     "Course" = "sd4413-devops-for-dev"
+   }
+ }

```

Results:

- VPC:

The screenshot shows the AWS Management Console interface for the VPC dashboard. The left sidebar contains navigation links for various services. The main content area displays 'Your VPCs' with a table showing one VPC. Below the table, the details for the selected VPC are shown.

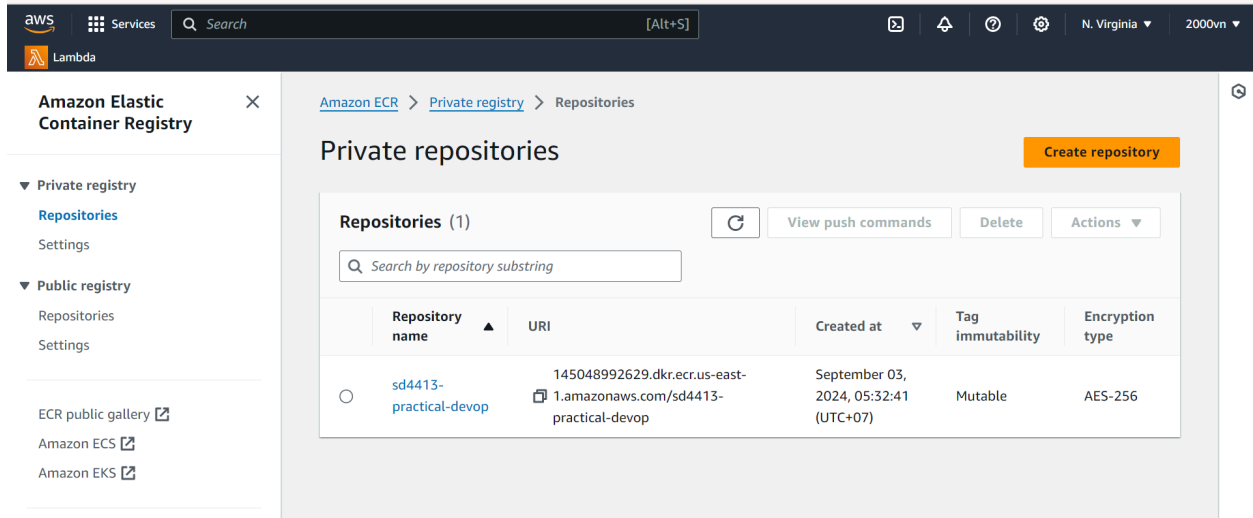
Name	VPC ID	State	IPv4 CIDR	IPv6 CIDR
No VPCs found in this Region				

vpc-0b89df8a51862c692 / my-vpc

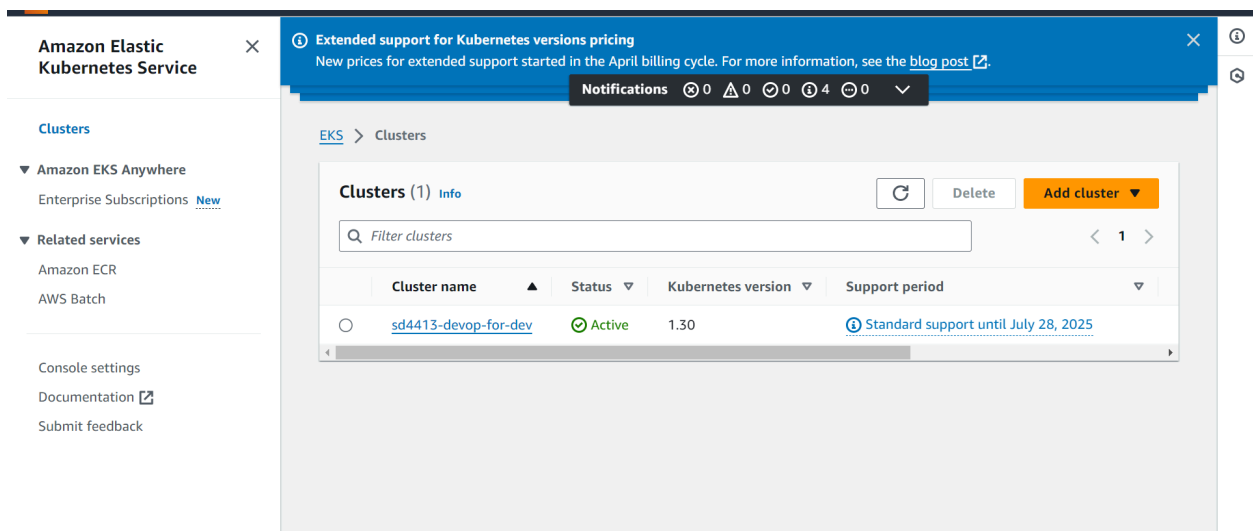
Details | Resource map | CIDRs | Flow logs | Tags | Integrations

Details			
VPC ID	State	DNS hostnames	DNS resolution
vpc-0b89df8a51862c692	Available	Enabled	Enabled

- ECR



• EKS



3. Setup Jenkins pipeline for CI/CD

• Check if Jenkins is installed:

Open a terminal and run the following command:

```
jenkins --version
```

```
vuongnguyenv@VNNOT02236:~/my-assignment/sd4413_aws_infrastructure$ jenkins --version
2.462.1
```

If the command is not found, Jenkins might not be installed, or the system's **PATH** may not include Jenkins.

- **Check if Jenkins is running:**

Jenkins usually runs as a service. You can check if it's running by using:

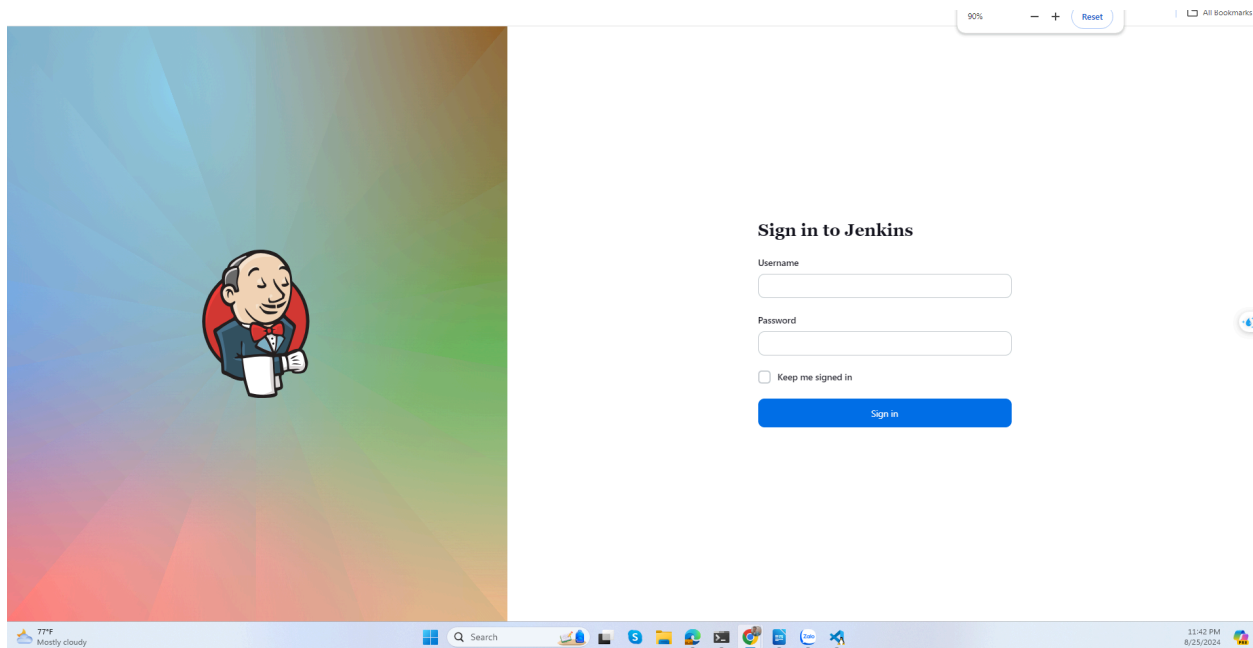
```
sudo systemctl status jenkins
```

The status should be **active (running)** if Jenkins is up.

```
[sudo] password for vuongnguyenv:
● jenkins.service - Jenkins Continuous Integration Server
   Loaded: loaded (/lib/systemd/system/jenkins.service; enabled; vendor preset: enabled)
   Active: active (running) since Sun 2024-08-25 19:08:37 +07; 4h 33min ago
     Main PID: 11906 (java)
        Tasks: 58 (limit: 9351)
       Memory: 1.0G
      CGroup: /system.slice/jenkins.service
              └─11906 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/java/jenkins.war --webroot=/var/cache/jer
```

- **Access Jenkins in a Browser:**

```
http://172.25.136.187:8080/login?from=%2F
```

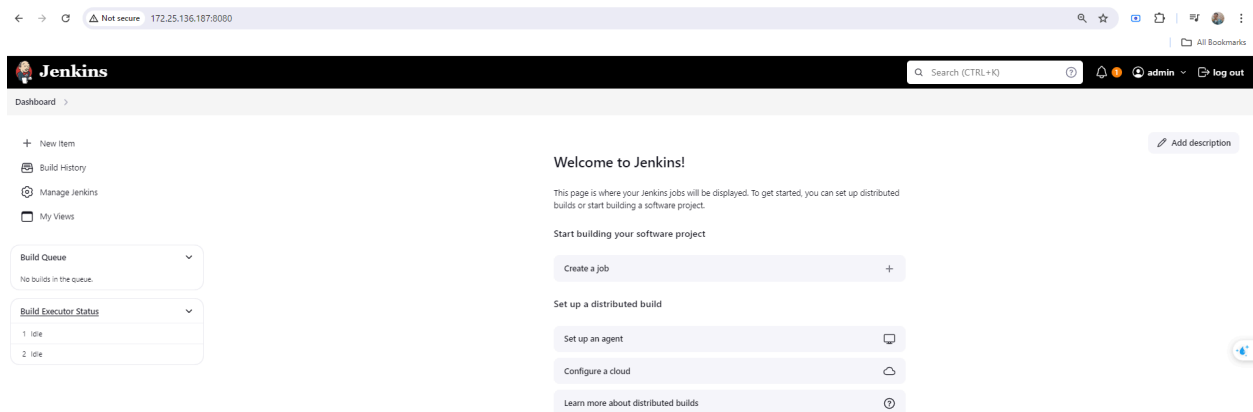


At the first time:

```
sudo cat /var/lib/jenkins/secrets/initialAdminPassword
```

for get password with default username is: **admin**

After login:

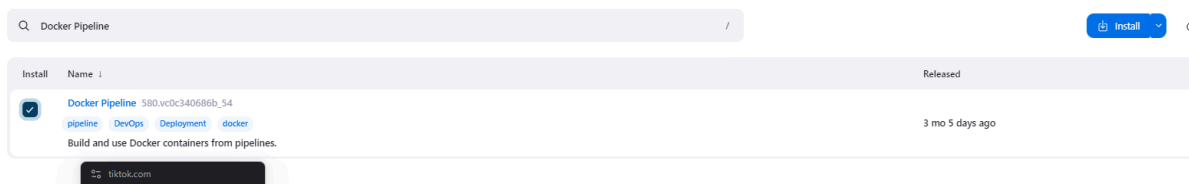


We should install some plugins:

1. Go to **Manage Jenkins > Manage Plugins**. Install some plugin listed below:

- **Pipeline Plugin:** Provides Jenkins support for building Docker images and running Docker containers in pipelines.

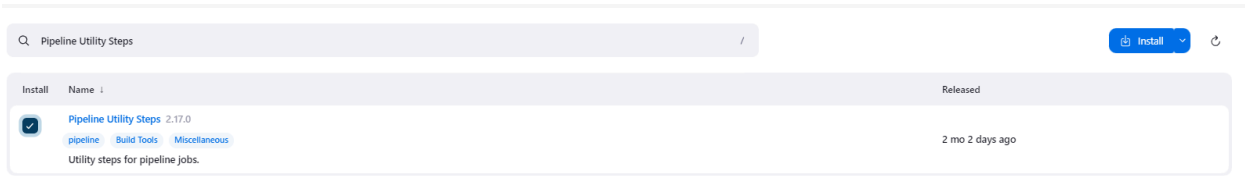
1. Go to the **Available** tab.
2. Search for **Docker Pipeline**.
3. Check the box next to **Docker Pipeline**.
4. Click **Install**



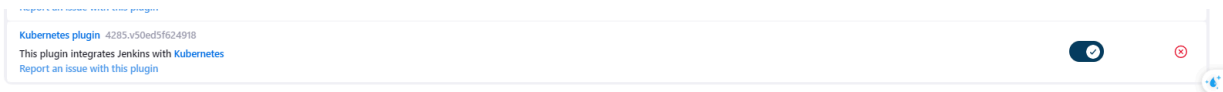
- **Pipeline Utility Steps Plugin:** Provides utility steps for Jenkins pipelines, such as reading and writing files.

1. Go to the **Available** tab.
2. Search for **Pipeline Utility Steps**.

3. Check the box next to **Pipeline Utility Steps**.
4. Click **Install**

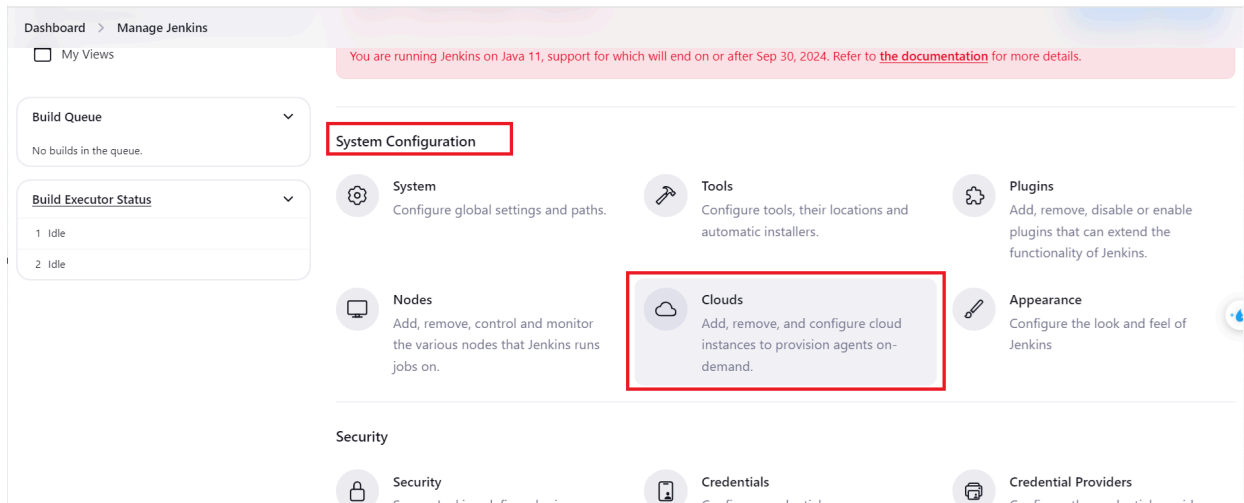


- **Kubernetes Plugin:** Provides integration with Kubernetes for Jenkins.
 1. Go to the **Available** tab.
 2. Search for **Kubernetes**.
 3. Check the box next to **Kubernetes**.
 4. Click **Install**

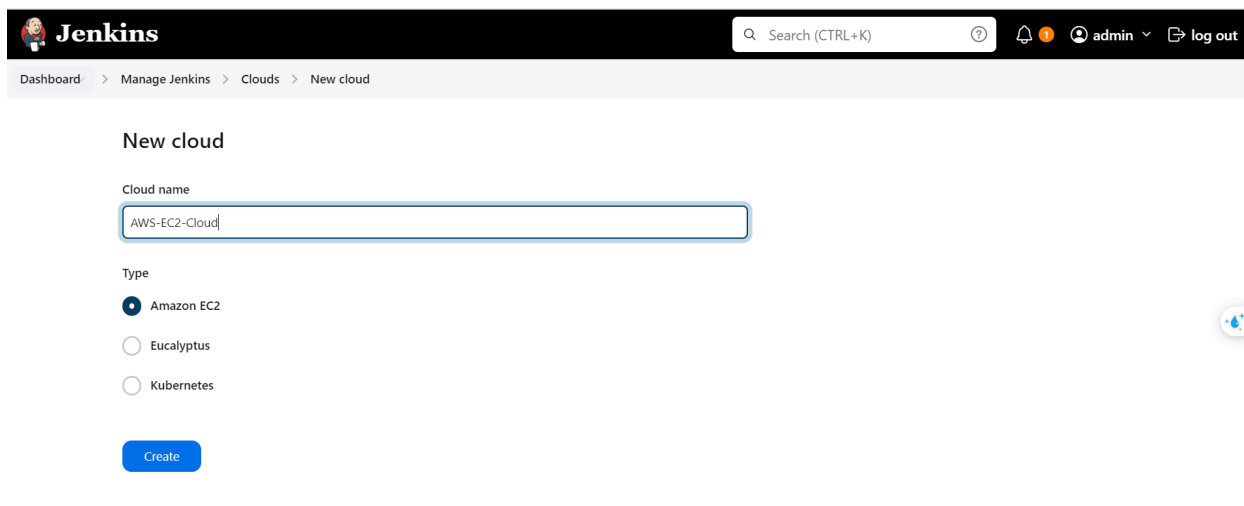


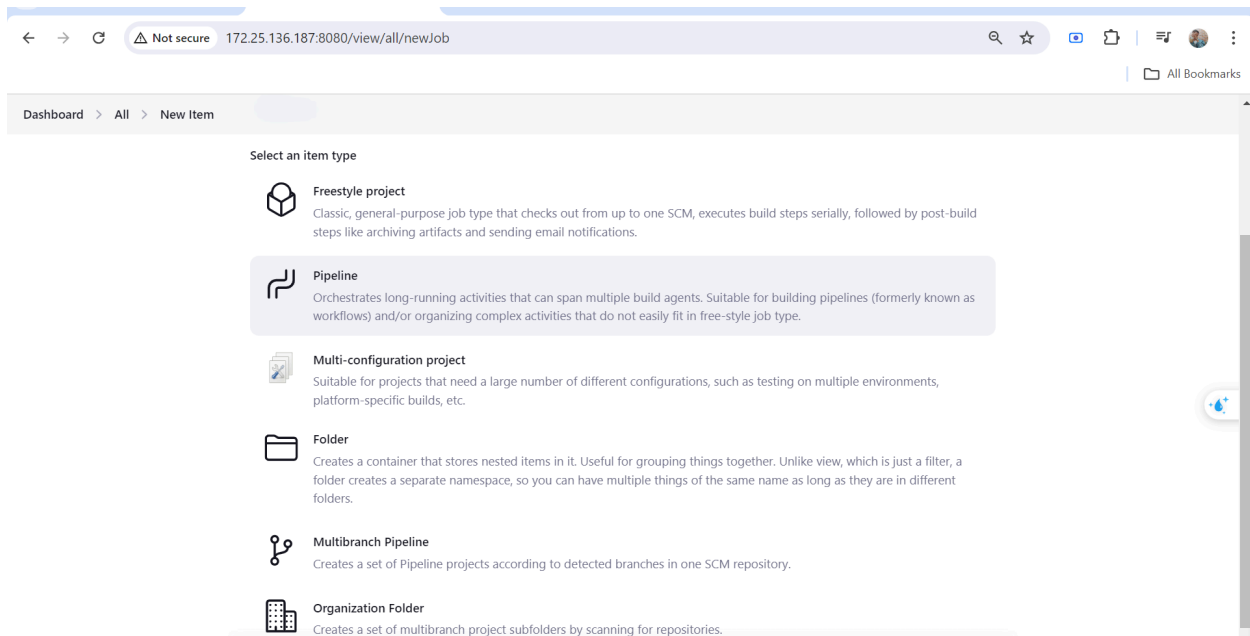
All plugins already installed. We can check at tab **Installed Plugins** .

- **Configure Amazon EC2 Plugin**
 1. **Go to Manage Jenkins > Configure System.**
 2. Scroll down to the **Cloud** section and click **Add a new cloud.**



3. Select **Amazon EC2** from the drop-down menu.



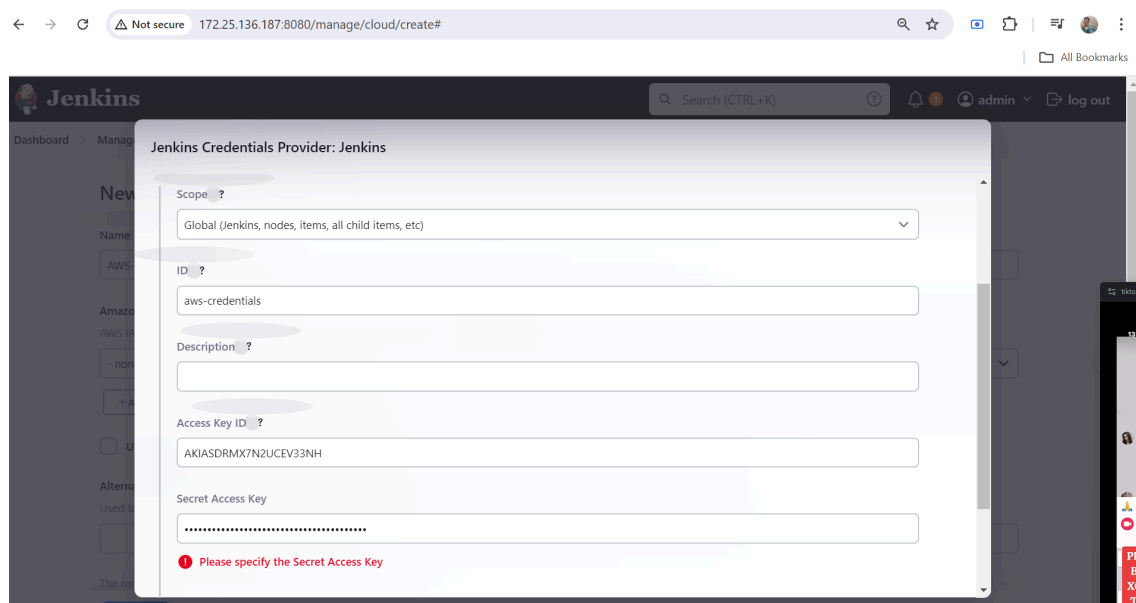


• Configure EC2 Settings

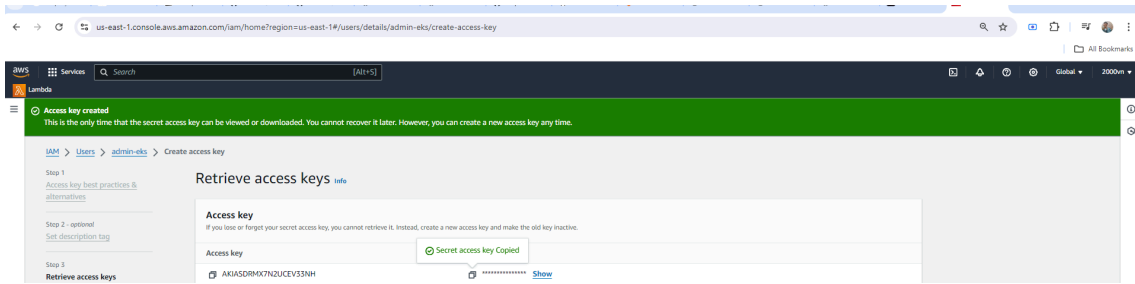
Fill in the EC2 cloud configuration settings:

◦ **AWS Credentials:**

- Click **Add** to enter your AWS credentials and we need to provide your AWS Access Key ID and Secret Access Key



- which it already existing AWS user:



- After adding credentials, select them from the drop-down menu.
- **Github Credentials:**

- Generate a New SSH Key

```
ssh-keygen -t rsa -b 4096 -C "nguyenvuongs2000@gmail.com"
```

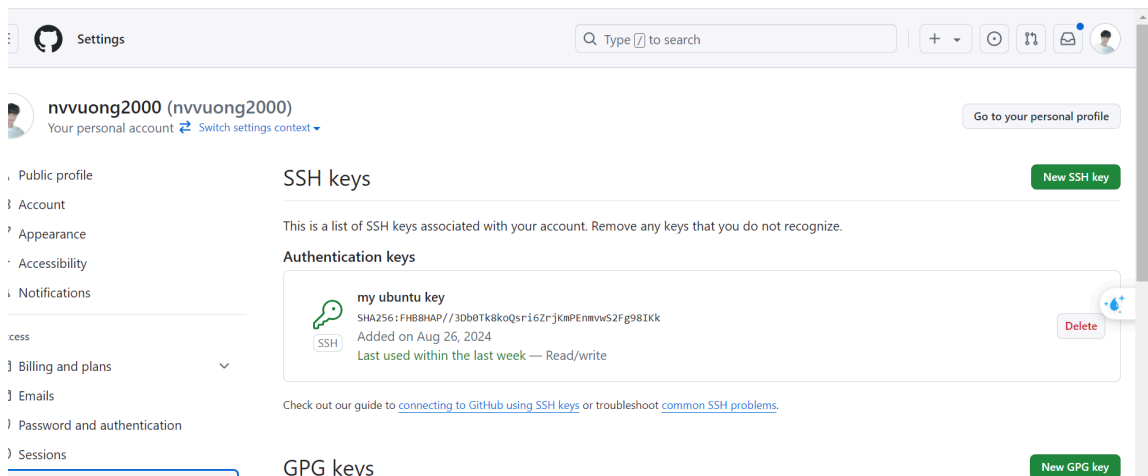
- Start the SSH agent:

```
eval "$(ssh-agent -s)"
```

- Then add your SSH key:

```
ssh-add ~/.ssh/id_ed25519
```

Now, go to GitHub SSH settings and click "New SSH key". Paste your key into the "Key" field, give it a title, and click "Add SSH key".



- Test SSH Connection

```
ssh -T git@github.com
```

```
76832f0b..588cfb7 master -> master
nvuongnguyenv@VNN0T02236:~/my-assignment/sd4413-ma$ ssh -T git@github.com
Hi nvvuong2000! You've successfully authenticated, but GitHub does not provide shell access.
```

- Back to UI. Fill in some fields required for build

Configuration

Definition

Pipeline script from SCM

SCM ?

Git

Repositories ?

Repository URL ?

https://github.com/nvvuong2000/sd4413_msa.git

Credentials ?

Branches to build ?

Branch Specifier (blank for 'any') ?

*/master

Add Branch

Repository browser ?

(Auto)

- Click save

Dashboard > sd-4413-jenkin-cicd >

Status

</> Changes

▶ Build Now

⚙️ Configure

🗑️ Delete Pipeline

🐱 GitHub

📁 Stages

✎ Rename

❓ Pipeline Syntax

✅ sd-4413-jenkin-cicd

Permalinks

- [Last build \(#15\), 1 hr 50 min ago](#)
- [Last stable build \(#15\), 1 hr 50 min ago](#)
- [Last successful build \(#15\), 1 hr 50 min ago](#)
- [Last failed build \(#14\), 2 hr 31 min ago](#)
- [Last unsuccessful build \(#14\), 2 hr 31 min ago](#)
- [Last completed build \(#15\), 1 hr 50 min ago](#)

☁️ Build History

trend ▼

← → ↻ ⚠️ Not secure 172.25.136.187:8080/job/sd-4413-jenkin-cicd/15/ 🔍 ☆ 📄 🗑️ 🔄 👤 ⋮

All Bookmarks

Jenkins

Search (CTRL+K) 🔍

🔔

👤 admin ▼

🚪 log out

Dashboard > sd-4413-jenkin-cicd > #15

Status

</> Changes

📄 Console Output

✎ Edit Build Information

🗑️ Delete build '#15'

⌚ Timings

🔍 Git Build Data

🔗 Pipeline Overview

📄 Pipeline Console

🔄 Restart from Stage

🔄 Replay

📋 Pipeline Steps

✅ Build #15 (Aug 26, 2024, 3:34:34 PM)

Keep this build forever

✎ Add description

Started 1 hr 52 min ago

Took 3 min 51 sec

</> Changes

1. fix jenkins file ([commit: 588cfb7](#)) ([details](#) / [githubweb](#))

🕒 Started by user [admin](#)

🕒 This run spent:

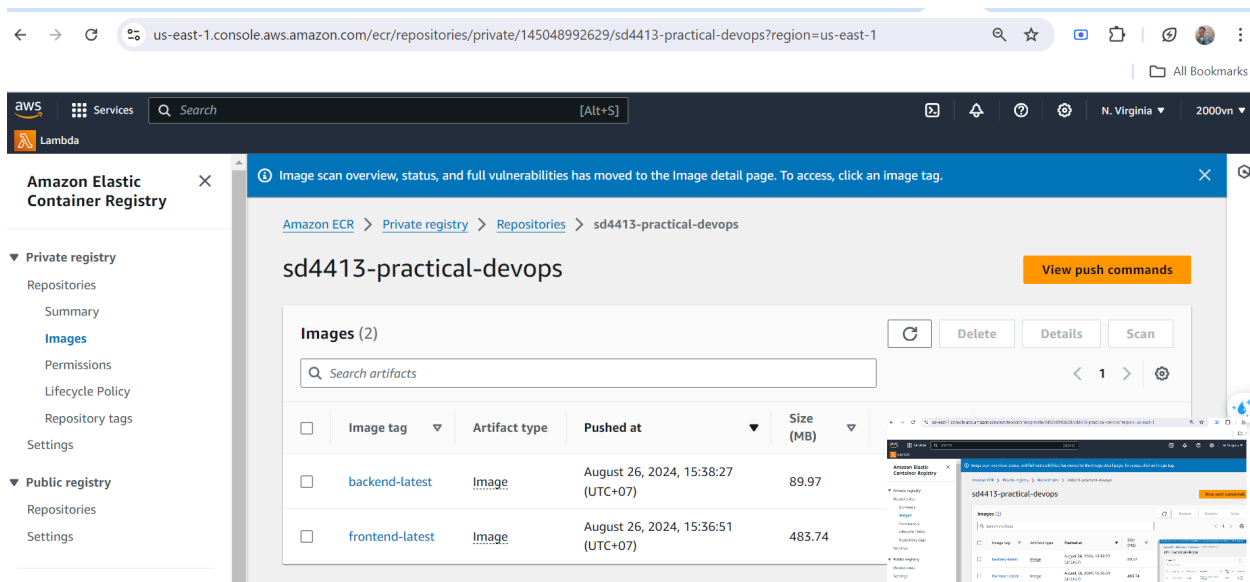
- 98 ms waiting;
- 3 min 51 sec build duration;
- 3 min 51 sec total from scheduled to completion.

🔍 **Revision:** 588cfb7733d0225e5ea38c8b061d52a46fd6d4a4

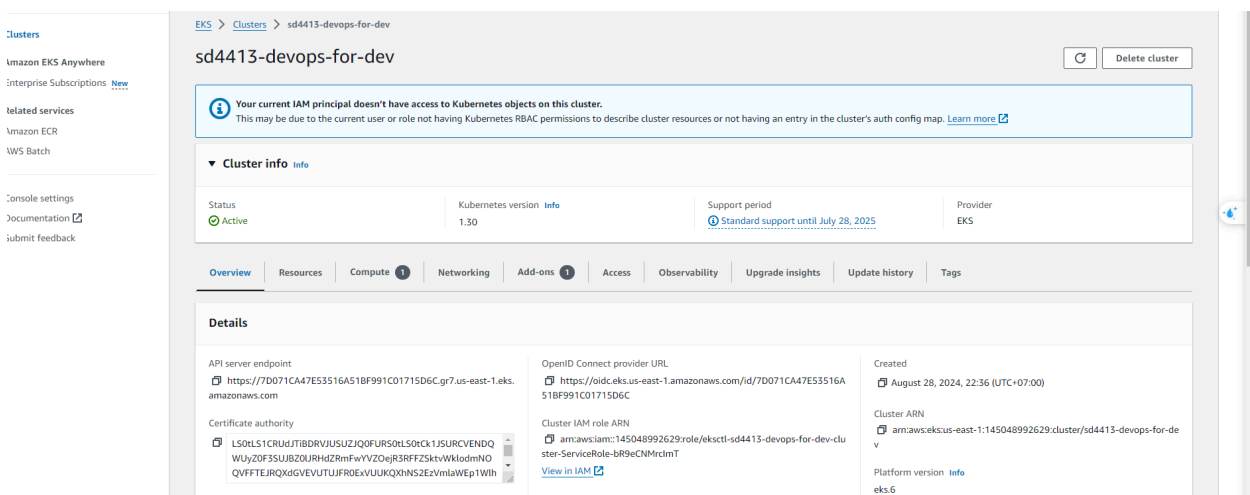
🔍 **Repository:** https://github.com/nvvuong2000/sd4413_msa.git

• [refs/remotes/origin/master](#)

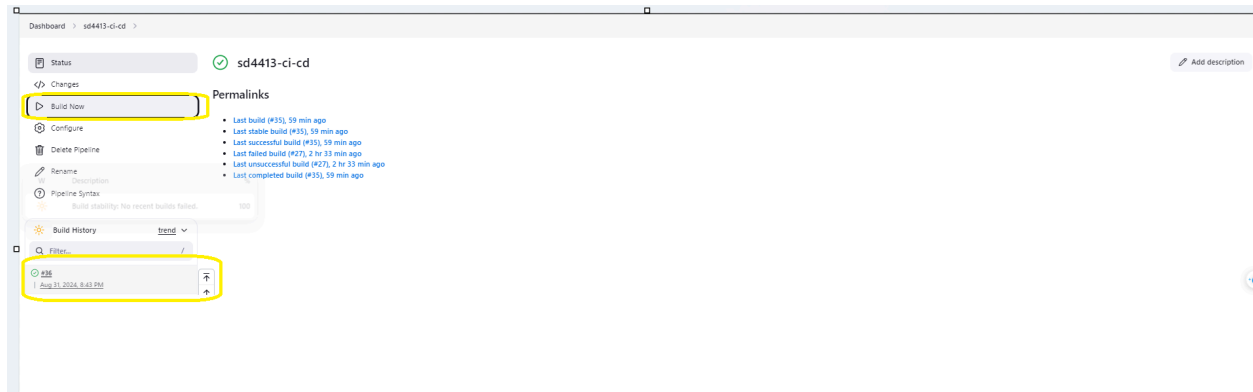
- Check at AWS ECR we can see 2 images have created successfully.



• EKS:



After completing the setup, I click on Deploy:



After deploy successfully. i can check pods and services:

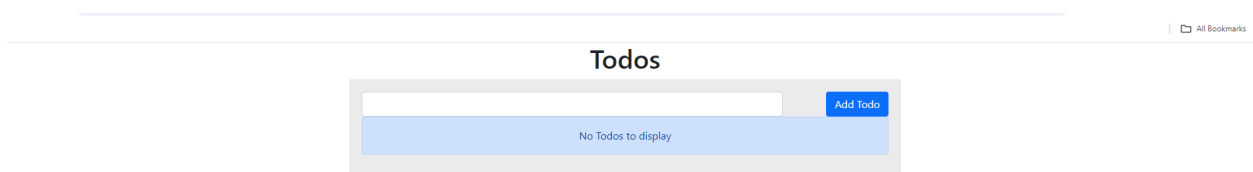
```

vuongnguyenv@VNNOT02236:~/learning/assignment/SD4413_msa$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
backend-787cf6d8-nzkzt              1/1     Running   0           48s
frontend-69f774599f-jcf6k          1/1     Running   1 (2s ago)  48s
mongo-555d8cdb8-rkbvx              1/1     Running   0           48s
vuongnguyenv@VNNOT02236:~/learning/assignment/SD4413_msa$ kubectl get services
NAME          TYPE        CLUSTER-IP    EXTERNAL-IP  PORT(S)    AGE
backend       ClusterIP   10.100.39.173 <none>       3000/TCP    2d20h
frontend      ClusterIP   10.100.24.78  <none>       3000/TCP    2d20h
kubernetes    ClusterIP   10.100.0.1    <none>       443/TCP     2d21h
mongo         ClusterIP   10.100.164.52 <none>       27017/TCP   113m
vuongnguyenv@VNNOT02236:~/learning/assignment/SD4413_msa$ kubectl get svc
NAME          TYPE        CLUSTER-IP    EXTERNAL-IP  PORT(S)    AGE
backend       LoadBalancer 10.100.39.173 ade9b8644528344f088851c6676dad66-494115400.us-east-1.elb.amazonaws.com 3000:31568/TCP 2d21h
frontend      LoadBalancer 10.100.24.78  a637835af732648c2b9aad2325b98cb5-1688344813.us-east-1.elb.amazonaws.com 3000:31188/TCP 2d21h
kubernetes    ClusterIP   10.100.0.1    <none>       443/TCP     2d21h
mongo         ClusterIP   10.100.164.52 <none>       27017/TCP   113m

```

I use **LoadBalancer** to expose urls to externally. This will provision an AWS Elastic Load Balancer (ELB) and provide an external IP or DNS name.

After adjust 2 files .yaml. I can see 2 urls have been created successfully



```

vuongnguyenv@VNNOT02236:~/learning/assignment/SD4413_msa$ kubectl get svc
NAME          TYPE        CLUSTER-IP    EXTERNAL-IP  PORT(S)    AGE
backend       LoadBalancer 10.100.39.173 ade9b8644528344f088851c6676dad66-494115400.us-east-1.elb.amazonaws.com 3000:31568/TCP 2d21h
frontend      LoadBalancer 10.100.24.78  a637835af732648c2b9aad2325b98cb5-1688344813.us-east-1.elb.amazonaws.com 3000:31188/TCP 2d21h
kubernetes    ClusterIP   10.100.0.1    <none>       443/TCP     2d21h
mongo         ClusterIP   10.100.164.52 <none>       27017/TCP   113m

```

4. Install ArgoCD

```
vuongnguyenv@VNNOT02236:~/Learning/assignment/SD4413_msa$ kubectl create namespace argocd
kubectl apply -n argocd -f https://raw.githubusercontent.com/argoproj/argo-cd/stable/manifests/install.yaml
namespace/argocd created
customresourcedefinition.apiextensions.k8s.io/applications.argoproj.io created
customresourcedefinition.apiextensions.k8s.io/applicationsets.argoproj.io created
customresourcedefinition.apiextensions.k8s.io/appprojects.argoproj.io created
serviceaccount/argocd-application-controller created
serviceaccount/argocd-applicationset-controller created
serviceaccount/argocd-dex-server created
serviceaccount/argocd-notifications-controller created
serviceaccount/argocd-redis created
serviceaccount/argocd-repo-server created
serviceaccount/argocd-server created
role.rbac.authorization.k8s.io/argocd-application-controller created
role.rbac.authorization.k8s.io/argocd-applicationset-controller created
role.rbac.authorization.k8s.io/argocd-dex-server created
role.rbac.authorization.k8s.io/argocd-notifications-controller created
role.rbac.authorization.k8s.io/argocd-redis created
role.rbac.authorization.k8s.io/argocd-server created
clusterrole.rbac.authorization.k8s.io/argocd-application-controller created
clusterrole.rbac.authorization.k8s.io/argocd-applicationset-controller created
clusterrole.rbac.authorization.k8s.io/argocd-server created
rolebinding.rbac.authorization.k8s.io/argocd-application-controller created
rolebinding.rbac.authorization.k8s.io/argocd-applicationset-controller created
rolebinding.rbac.authorization.k8s.io/argocd-dex-server created
rolebinding.rbac.authorization.k8s.io/argocd-notifications-controller created
rolebinding.rbac.authorization.k8s.io/argocd-redis created
rolebinding.rbac.authorization.k8s.io/argocd-server created
clusterrolebinding.rbac.authorization.k8s.io/argocd-application-controller created
clusterrolebinding.rbac.authorization.k8s.io/argocd-applicationset-controller created
clusterrolebinding.rbac.authorization.k8s.io/argocd-server created
configmap/argocd-cm created
configmap/argocd-cmd-params-cm created
configmap/argocd-gpg-keys-cm created
configmap/argocd-notifications-cm created
configmap/argocd-rbac-cm created
configmap/argocd-ssh-known-hosts-cm created
configmap/argocd-tls-certs-cm created
secret/argocd-notifications-secret created
secret/argocd-secret created
service/argocd-applicationset-controller created
service/argocd-dex-server created
```

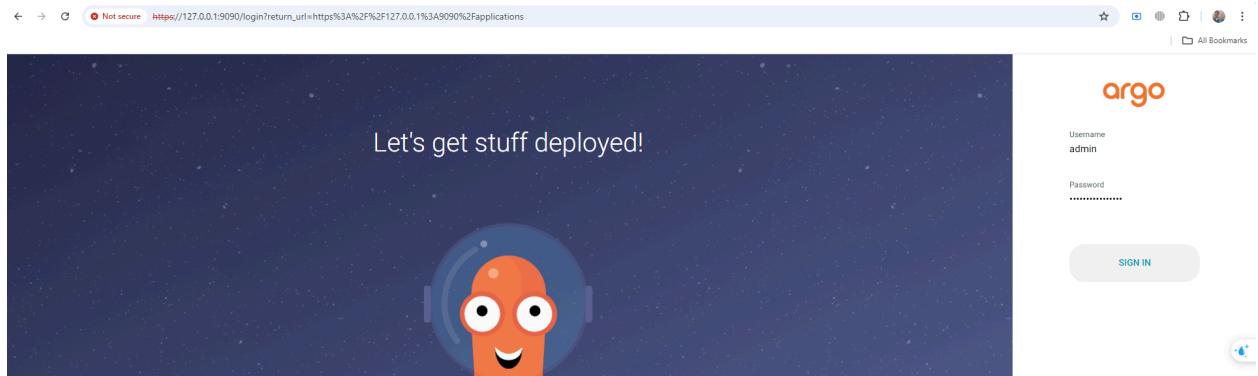
Forced to use a different port because Jenkins was already using port 8080.

```
networkpolicy.networking.k8s.io/argocd-server networkpolicy created
vuongnguyenv@VNNOT02236:~/Learning/assignment/SD4413_msa$ kubectl port-forward svc/argocd-server -n argocd 9090:443
Forwarding from 127.0.0.1:9090 -> 8080
Forwarding from [::1]:9090 -> 8080
```

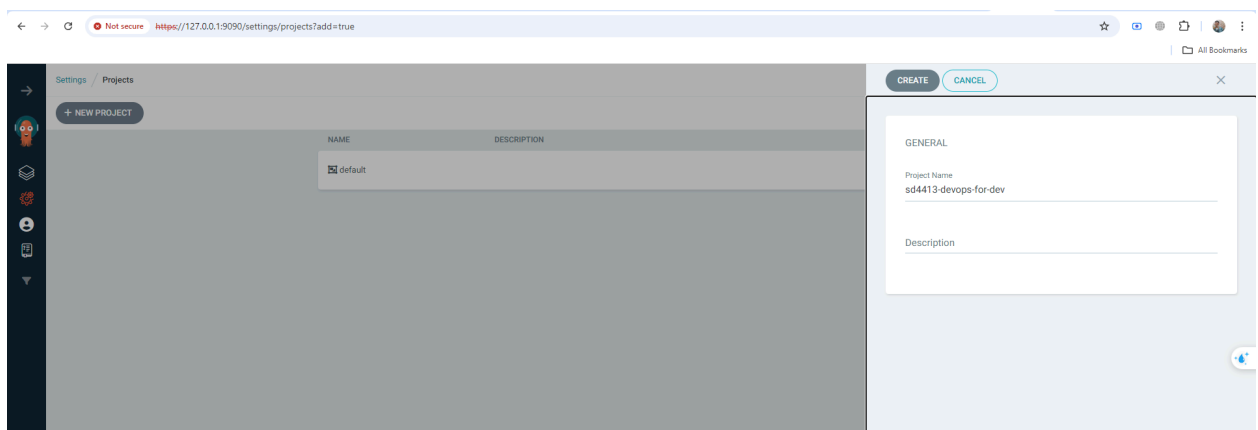
- Open your browser and navigate to <https://127.0.0.1:9090/>.
- **Login to Argo CD:**

By default, the username is `admin`, and the password is the name of the Argo CD server pod. Retrieve the pod name and log in:

```
vuongnguyenv@VNNOT02236:~/Learning/assignment/SD4413_msa$ kubectl -n argocd get secret argocd-initial-admin-secret -o jsonpath="{.data.password}" | base64 -d; echo
KEi4gBXqI2sC0w8T
vuongnguyenv@VNNOT02236:~/Learning/assignment/SD4413_msa$
```



"In the left navigation, click on 'Project'. Create a new project with the name...



Fill in data for some sections

Source Repositories:

SOURCE REPOSITORIES ⓘ

SAVE
CANCEL

*

ADD SOURCE

SCOPED REPOSITORIES ⓘ

Project has no source repositories

DESTINATIONS ⓘ

SAVE
CANCEL

Server	Name	Namespace
*	*	*

ADD DESTINATION

Destinations, Cluster resource allow list, Name spaces resources allow list:

CLUSTER RESOURCE ALLOW LIST ⓘ

SAVE
CANCEL

Kind	Group
*	*

ADD RESOURCE

CLUSTER RESOURCE DENY LIST ⓘ

Kind	Group
------	-------

ADD RESOURCE

NAMESPACE RESOURCE ALLOW LIST ⓘ

Kind	Group
*	*

ADD RESOURCE

NAMESPACE RESOURCE DENY LIST ⓘ

Kind	Group
------	-------

ADD RESOURCE

Back to [Home](#) . Click create new application and fill in data

GENERAL

EDIT AS YAML

Application Name

sd4413-devops-for-dev

Project Name

sd4413-devops-for-dev

SYNC POLICY

Manual

☐ SET DELETION FINALIZER

SYNC OPTIONS

☐ SKIP SCHEMA VALIDATION
☐ PRUNE LAST
☐ RESPECT IGNORE DIFFERENCES

☐ AUTO-CREATE NAMESPACE
☐ APPLY OUT OF SYNC ONLY
☐ SERVER-SIDE APPLY

PRUNE PROPAGATION POLICY: foreground

☐ REPLACE
☐ RETRY

SOURCE

Repository URL

https://github.com/nvnuong2000/sd4413_msa.git

GIT

Revision

master

Branches

Path

./k8s/

DESTINATION

Cluster URL

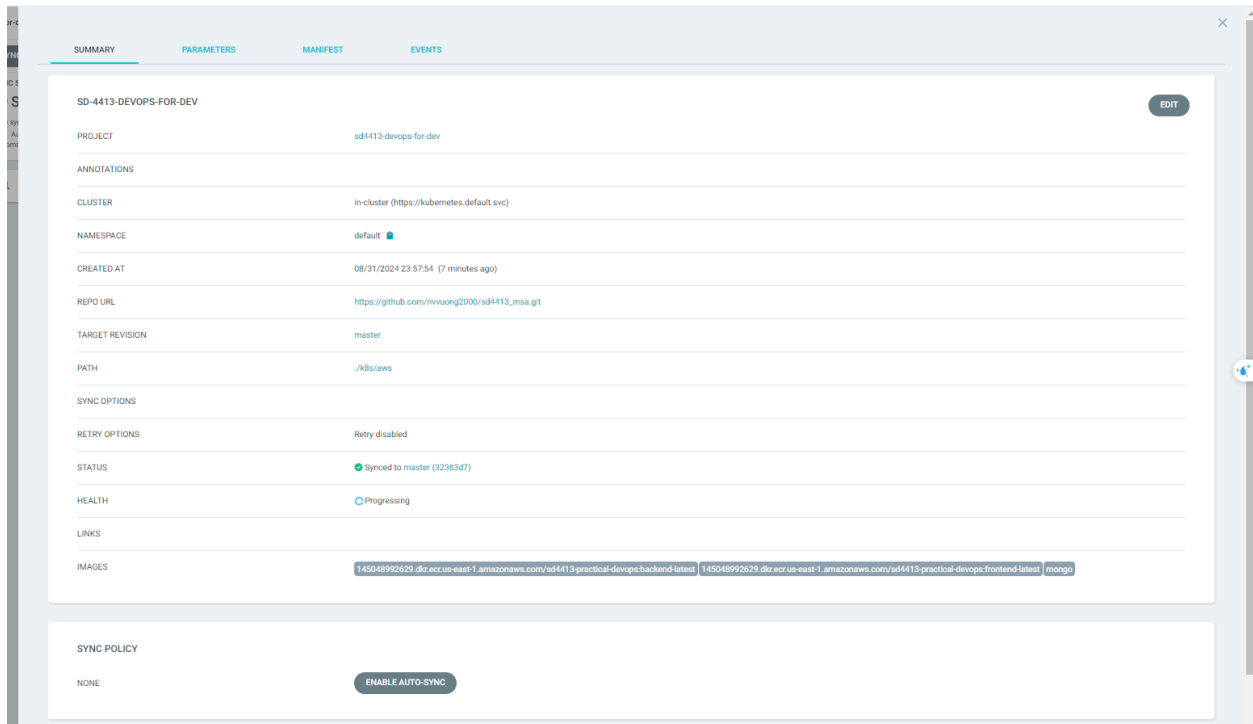
https://kubernetes.default.svc

URL

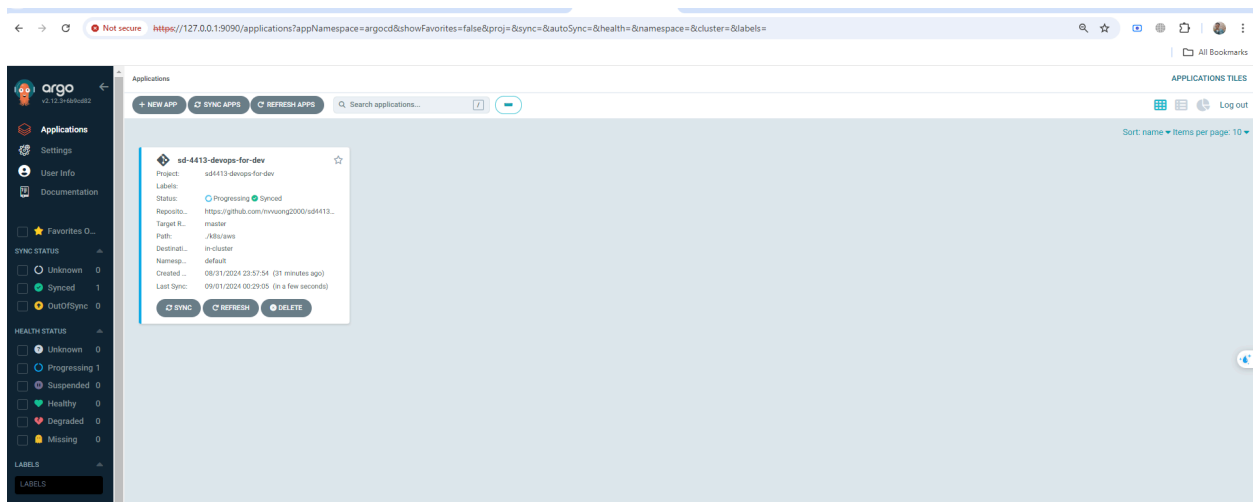
Namespace

sd4413-devops-for-dev-argo

Preview project's setting. And click create.



Create new app successfully.



Change replicas to 2 and click to refresh. We can see currently status is **OutOfSync**

Commit

The screenshot shows a Git commit interface. At the top, the commit message is "change replicas" and the branch is "master". Below this, it says "nvvuong2000 committed 4 minutes ago". The diff shows a change in the file "k8s/awx/bockend.yml". The change is a deletion of a line: "replicas: 1" on line 23. The diff is color-coded: blue for additions, red for deletions, and green for context. The file path is "k8s/awx/bockend.yml".

The screenshot shows the Argo CD web interface. The browser address bar shows the URL "https://127.0.0.1:9090/applications?showFavorites=false&proj=&sync=&autoSync=&health=&namespace=&cluster=&labels=". The interface has a sidebar with navigation links: Applications, Settings, User Info, and Documentation. The main content area shows a list of applications. The application "sd-4413-devops-for-dev" is selected, and its details are shown in a modal. The details include: Project: sd4413-devops-for-dev, Label: sd-4413-devops-for-dev, Status: Progressing, OutOfSync, Repository: https://github.com/nvuuong2000/sd4413..., Target R: master, Path: /k8s/awx, Destination: in-cluster, Namespace: default, Created: 08/31/2024 23:57:54 (26 minutes ago), Last Sync: 09/01/2024 00:02:55 (21 minutes ago). At the bottom of the modal, there are buttons for "SYNC", "REFRESH", and "DELETE".

Click **synchronize** for sync any changes.

SYNCHRONIZE

CANCEL

✕

Synchronizing application manifests from
https://github.com/nvvuong2000/sd4413_msa.git

Revision
master

☐ PRUNE ☐ DRY RUN ☐ APPLY ONLY ☐ FORCE

SYNC OPTIONS

☐ SKIP SCHEMA VALIDATION ☐ AUTO-CREATE NAMESPACE

☐ PRUNE LAST ☐ APPLY OUT OF SYNC ONLY

☐ RESPECT IGNORE DIFFERENCES ☐ SERVER-SIDE APPLY

PRUNE PROPAGATION POLICY: **foreground** ▾

☐ REPLACE ⚠

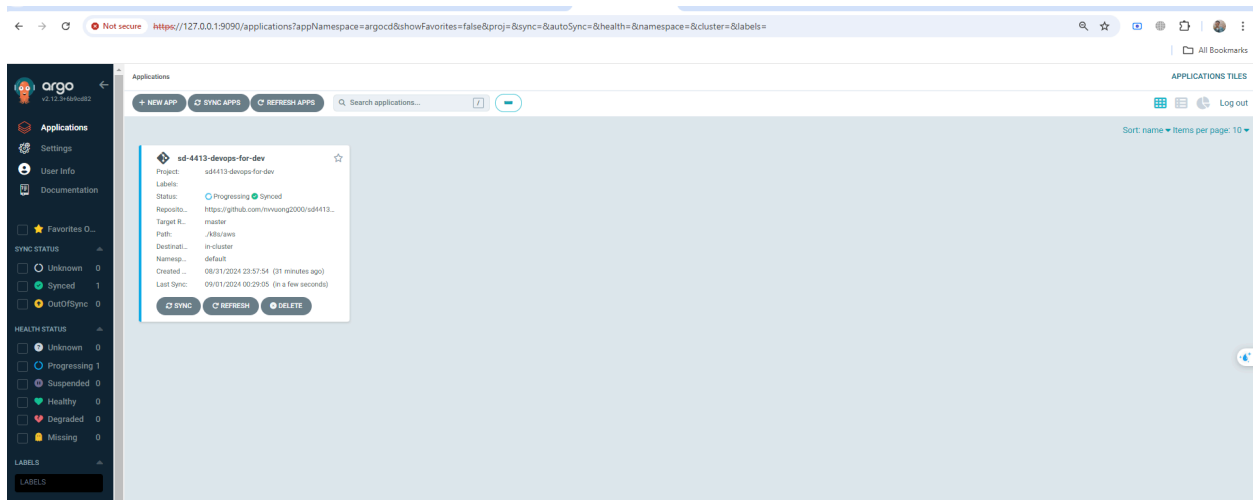
☐ RETRY

SYNCHRONIZE RESOURCES: all / out of sync / non 

- ☒ /CONFIGMAP/DEFAULT/MONGO-CONFIG ✓
- ☒ /SECRET/DEFAULT/MONGO-SECRETS ✓
- ☒ /SERVICE/DEFAULT/BACKEND ✓
- ☒ /SERVICE/DEFAULT/FRONTEND ✓
- ☒ /SERVICE/DEFAULT/MONGO ✓
- ☒ APPS/DEPLOYMENT/DEFAULT/BACKEND ✓
- ☒ APPS/DEPLOYMENT/DEFAULT/FRONTEND ✓
- ☒ APPS/DEPLOYMENT/DEFAULT/MONGO ✓

12:39 AM 

After sync currently status is **Synced**



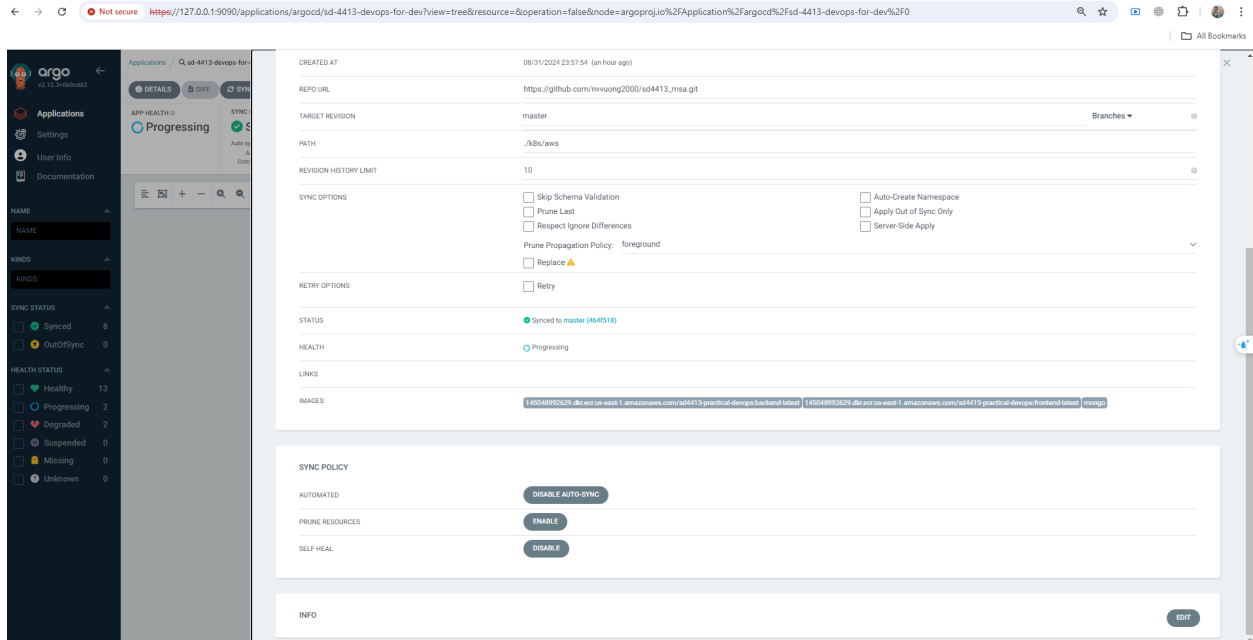
```
vuongnguyenv@VN0T82236:~/Learning/assignment/SD4413_msa$ kubectl scale --replicas=3 deployment backend -n default
deployment.apps/backend scaled
vuongnguyenv@VN0T82236:~/Learning/assignment/SD4413_msa$ kubectl get deployment backend -n default -o wide
NAME READY UP-TO-DATE AVAILABLE AGE CONTAINERS IMAGES SELECTOR
backend 3/3 3 3 3d1h backend 145948992629.dkr.ecr.us-east-1.amazonaws.com/sd4413-practical-devops:backend-latest app=backend
vuongnguyenv@VN0T82236:~/Learning/assignment/SD4413_msa$
```

Avoid manual changes in my cluster.

We can use feature **enabling self-heal**

Go to the **Argo CD dashboard** and click on the application. In the Application screen click the top left **App** ⇒ **Details**

Scroll down and find the **Sync policy section**. Click on the **Self-heal** button to enable



We can try to change replicas at backed service to **3**.

```

vuongnguyenv@VNNOT02236:~/Learning/assignment/SD4413_msa$ kubectl scale --replicas=3 deployment backend -n default
deployment.apps/backend scaled
vuongnguyenv@VNNOT02236:~/Learning/assignment/SD4413_msa$ kubectl get deployment backend -n default

```

After that we can check again:

```

vuongnguyenv@VNNOT02236:~/Learning/assignment/SD4413_msa$ kubectl get deployment backend -n default -o wide
NAME READY UP-TO-DATE AVAILABLE AGE CONTAINERS IMAGES SELECTOR
backend 2/2 2 2 3d1h backend 145048992629.dkr.ecr.us-east-1.amazonaws.com/sd4413-practical-devops:backend-latest app=backend

```

My application will always have **2** pods deployed. This is what the Git state says and therefore Argo CD automatically makes the cluster state the same.

5. Set up Prometheus and Grafana to monitor EKS resources and default EKS resource metric.

- Install helm

```

vuongnguyenv@VN0T02236:~/learning/assignment/SD4413_msa$ curl https://baltocdn.com/helm/signing.asc | sudo apt-key add -
sudo apt-get install apt-transport-https --yes
echo "deb https://baltocdn.com/helm/stable/debian/ all main" | sudo tee /etc/apt/sources.list.d/helm-stable-debian.list
sudo apt-get update
sudo apt-get install helm
[sudo] password for vuongnguyenv:
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left   Speed
100 1699 100 1699    0     0  5817      0 --:--:-- --:--:-- --:--:--  5818

Warning: apt-key is deprecated. Manage keyring files in trusted.gpg.d instead (see apt-key(8)).
OK
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
apt-transport-https is already the newest version (2.4.12).
0 upgraded, 0 newly installed, 0 to remove and 10 not upgraded.
deb https://baltocdn.com/helm/stable/debian/ all main
Hit:1 https://download.docker.com/linux/ubuntu jammy InRelease
Hit:2 https://packages.microsoft.com/repos/visualstudio stable InRelease
Hit:3 https://packages.microsoft.com/repos/code stable InRelease
Get:4 https://baltocdn.com/helm/stable/debian all InRelease [7652 B]
Ign:5 https://pkg.jenkins.io/debian-stable binary/ InRelease
Hit:6 https://pkg.jenkins.io/debian-stable binary/ Release
Get:8 https://baltocdn.com/helm/stable/debian all/main amd64 Packages [4360 B]
Hit:9 http://archive.ubuntu.com/ubuntu jammy InRelease
Get:11 http://security.ubuntu.com/ubuntu jammy-security InRelease [129 kB]
Get:12 http://archive.ubuntu.com/ubuntu jammy-updates InRelease [128 kB]
Get:13 https://packages.cloud.google.com/apt/kubernetes-xenial InRelease
Err:14 http://archive.ubuntu.com/ubuntu jammy-backports InRelease
404 Not Found [IP: 102.250.197.46 404]
Reading package lists... Done
W: https://download.docker.com/linux/ubuntu/dists/jammy/InRelease: Key is stored in legacy trusted.gpg keyring (/etc/apt/trusted.gpg), see the DEPRECATION section in apt-key(8) for details.
W: https://baltocdn.com/helm/stable/debian/dists/all/InRelease: Key is stored in legacy trusted.gpg keyring (/etc/apt/trusted.gpg), see the DEPRECATION section in apt-key(8) for details.
E: The repository 'https://apt.kubernetes.io/kubernetes-xenial Release' does not have a Release file.
N: Updating from such a repository can't be done securely, and is therefore disabled by default.
N: See apt-secure(8) manpage for repository creation and user configuration details.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  helm
0 upgraded, 1 newly installed, 0 to remove and 10 not upgraded.
Need to get 16.6 MB of archives.
After this operation, 52.5 MB of additional disk space will be used.
Get:1 https://baltocdn.com/helm/stable/debian all/main amd64 helm amd64 3.15.4-1 [16.6 MB]
Fetched 16.6 MB in 1s (15.0 MB/s)
Selecting previously unselected package helm.
(Reading database ... 55812 files and directories currently installed.)

```

- Check version

```

N: See apt-secure(8) manpage for repository creation and user configuration details.
vuongnguyenv@VN0T02236:~/learning/assignment/SD4413_msa$ helm version
version.BuildInfo{Version:"v3.15.4", GitCommit:"fa9e7b07d9d8debbb4306d72af76a383895aa8c4", GitTreeState:"clean", GoVersion:"go1.22.6"}
vuongnguyenv@VN0T02236:~/learning/assignment/SD4413_msa$

```

- Install `prometheus` , `grafana` with helm:

```

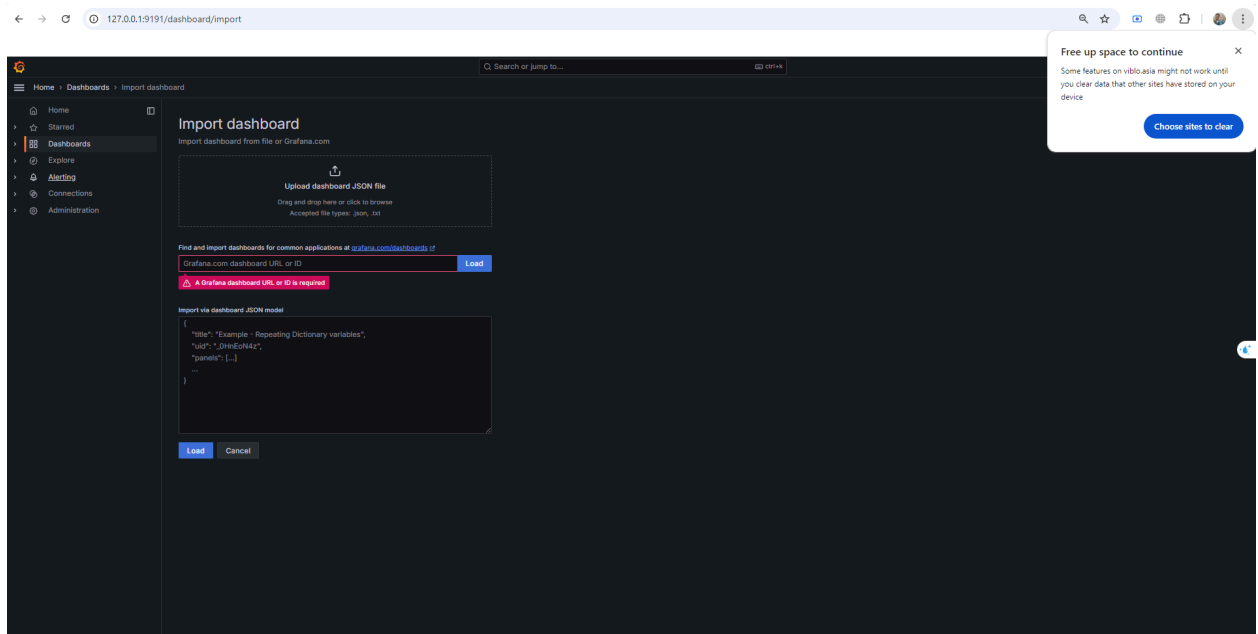
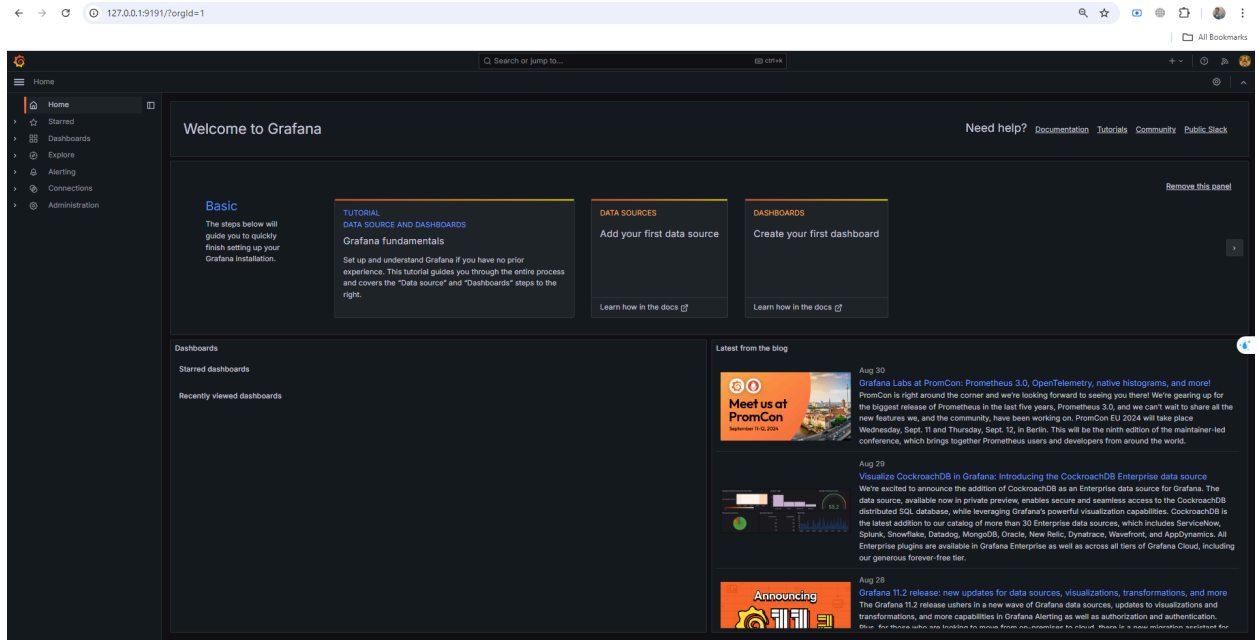
helm repo add prometheus-community https://prometheus-community.github.io/helm-charts
helm repo add stable https://charts.helm.sh/stable
helm repo update

```

```

vuongnguyenv@VN0T02236:~/learning/assignment/SD4413_msa$ helm repo add prometheus-community https://prometheus-community.github.io/helm-charts
helm repo add stable https://charts.helm.sh/stable
"prometheus-community" has been added to your repositories
"stable" has been added to your repositories
Hang tight while we grab the latest from your chart repositories...
...Successfully got an update from the "prometheus-community" chart repository
...Successfully got an update from the "stable" chart repository
Update Complete. Happy Helming!
vuongnguyenv@VN0T02236:~/learning/assignment/SD4413_msa$

```



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Nearly all default values exported by Prometheus node exporter graphed.

Only requires the default `job_name: node`, add as many targets as you need in `/etc/prometheus/prometheus.yml`.

```

- job_name: node
  static_configs:
    - targets: ['localhost:9100']

```

Recommended for prometheus-node-exporter the arguments `-collector.systemd -collector.processes` because the graph uses some of their metrics.

Since revision 16, for prometheus-node-exporter v0.18 or newer. Since revision 12, for prometheus-node-exporter v0.16 or newer.

Available on github: <https://github.com/rfmoz/grafana-dashboards.git>

2

Import the dashboard template

Copy ID to clipboard

or

Download JSON

Resources

Docs: Importing dashboards

Webinar: Getting started with Grafana dashboard design

Webinar: Building advanced Grafana dashboards

ID

1860

Datasource

Prometheus

Dependencies

Bar gauge 5.0.0 Gauge 5.0.0

Grafana 9.4.3 Stat 5.0.0

Time series 5.0.0

Published by

rfmoz

Last update

2024-05-22T16:07:35

Revisions

Revision	Description	Created	
37		2024-05-22T16:07:35	Download
36		2024-03-09T06:33:51	Download
35		2024-03-04T19:47:40	Download
34		2024-03-01T20:17:11	Download
33		2023-10-28T14:50:18	Download
32		2023-09-01T12:04:55	Download
31		2023-03-30T13:35:21	Download
30		2023-02-03T09:10:03	Download
29		2022-11-08T20:05:45	Download
28		2022-10-21T11:20:59	Download
27		2022-04-29T09:30:12	Download
26		2022-02-08T19:51:42	Download
25		2022-01-10T08:41:23	Download
24		2021-12-17T08:00:39	Download

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3:28 AM
9/1/2024

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Import dashboard

Import dashboard from file or Grafana.com

Importing dashboard from Grafana.com

Published by: rfmoz

Updated on: 2024-05-22 23:07:35

Options

Name: Node Exporter Full

Dashboards: [dropdown]

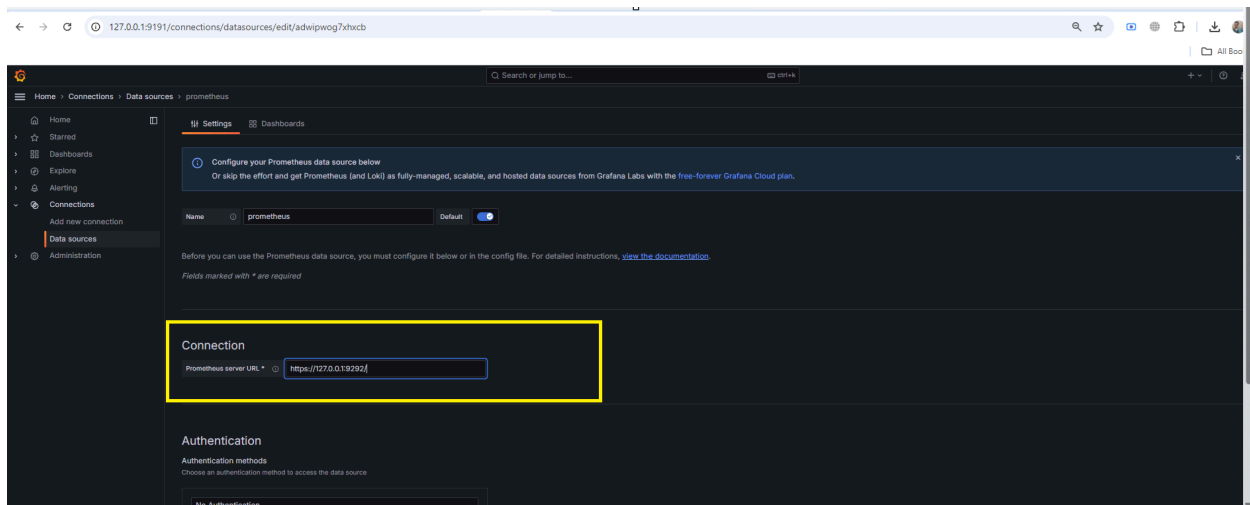
Unique Identifier (UID): [text area]

Prometheus: [search box]

No data sources found

Configure a new data source

Open advanced data source picker



Results:

